

for example, when the Moon passes in front of a star or when a planet (such as Jupiter) passes in front of one of its moons.

Oort Cloud (Oort–Öpik Cloud) A spherical distribution of trillions of icy planetesimals and cometary nuclei that surrounds the solar system and extends out to a radius of about 1.6 light-years from the Sun. It provides the reservoir from which long-period and “new” comets originate. Its existence was proposed in 1950 by Dutch astronomer Jan H. Oort (a similar idea had also been suggested by Estonian astronomer Ernst J. Öpik). See also *comet*, *planetesimal*.

open cluster A loose cluster of up to a few thousand stars that lies in or close to the plane of the Milky Way galaxy. Member stars of each cluster formed from the same cloud of gas and dust, and have closely similar ages and chemical compositions. Clusters of this kind are also known as **galactic clusters**. See also *globular cluster*.

open universe A universe in which the average density is less than the critical density that is needed to halt its expansion and which, therefore, will expand forever. See also *closed universe*, *flat universe*, *oscillating universe*.

opposition The position of a planet when it is exactly on the opposite side of Earth from the Sun. Its elongation is then 180°, and it is highest in the sky at midnight. See also *conjunction*, *elongation*.

optical double star see *double star*.

orbit The path of a body that is moving within the gravitational field of another. The orbit of a planet around a star or a satellite around a planet will normally be an ellipse or, exceptionally, a circle (a circle is a special case of an ellipse).

orbital period The period of time during which a body travels once around its orbit. The **sidereal orbital period** is the time taken by one body to revolve around another (for example, the Moon around Earth) measured relative to the background stars.

oscillating universe A universe that expands and contracts in a cyclic fashion. The collapse of such a universe at the end of one cycle triggers a new Big Bang that initiates the next cycle. See *closed universe*, *flat universe*, *open universe*.

P

parallax The apparent shift in position of an object when it is observed from different locations. **Stellar parallax** is the apparent shift in position of a relatively nearby star when viewed from different points on Earth's orbit. **Annual parallax** is the maximum angular displacement of a star from its mean position due to parallax. The greater the distance of a star, the smaller its parallax.

parhelic circle See *sun dog*.

parsec (pc) The distance at which a star would have an annual parallax of one second of arc (one second of angular measurement). One parsec is equivalent to 3.26 light-years, or 19,200 billion miles (30,900 billion km). See also *parallax*.

parselene See *sun dog*.

penumbra (1) The lighter, outer part of the shadow cast by an opaque body. An observer within the penumbra can see part of the illuminating source. See also *eclipse*. (2) The less dark and less cool outer region of a sunspot. See also *sunspot*, *umbra*.

perigee The point on its orbit at which a body that is revolving around Earth is at its closest to Earth. See also *apogee*.

perihelion The point on its orbit at which a planet, or other solar system body, is at its closest to the Sun.

phase The proportion of the visible hemisphere of the Moon or a planet that is illuminated by the Sun at any particular instant.

photon An individual package, or quantum, of electromagnetic energy, which may be envisaged as a “particle” of light. The shorter the wavelength of the radiation and higher the frequency, the greater the energy of the photon. See also *electromagnetic radiation*.

photosphere The thin, gaseous layer at the base of the solar atmosphere, from which the Sun's visible light is emitted and which corresponds to the visible surface of the Sun.

planet A body that is much less massive than a star, revolves around a star, and shines by reflecting that star's light. As a general guide, an orbiting body is considered to be a planet (rather than a brown dwarf) if its mass is less than about 13 times the mass of Jupiter. See also *brown dwarf star*.

planetary nebula A glowing shell of gas ejected by a star at a late stage in its evolution.

planetesimal One of the large number of small bodies, composed of rock or ice, that formed within the solar nebula and from which the planets were eventually assembled through the process of accretion.

plasma A completely ionized gas state of matter that consists of equal numbers of positively charged ions and negatively charged electrons. Plasmas usually have very high temperatures. Examples include the solar corona and solar wind, both of which consist predominantly of protons and electrons. See also *corona*, *solar wind*.

polar axis see *equatorial mounting*.

positron see *antiparticle*.

precession A slow change in the orientation of a rotating body's axis caused by the gravitational influence of neighboring bodies. Earth's axis precesses around in a conical pattern over a period of 25,800 years.

prominence A flamelike plume of gas that follows magnetic field lines in the solar atmosphere. An **active** or **eruptive prominence** undergoes rapid changes, whereas a **quiescent prominence** remains suspended in the solar atmosphere for a prolonged period.

proper motion The angular rate at which a star changes its observed position on the celestial sphere. **Annual proper motion** is the angle (seldom more than a small fraction of 1 second of angular measurement) through which a star appears to shift in the course of one year.

protogalaxy A progenitor of a normal galaxy. The building blocks from which galaxies were assembled through a process of collisions and mergers, protogalaxies are believed to have formed a few hundred million years after the Big Bang when clouds of gas collapsed under the action of gravity.

proton An elementary particle, composed of three quarks, that has a positive electrical charge and is a constituent of every atomic nucleus. See also *atom*.

proton–proton chain (pp chain)

A sequence of reactions that fuse together hydrogen nuclei (protons) to create helium nuclei. The net result of the process is to convert four protons into one helium nucleus, which consists of two protons and two neutrons. The proton–proton reaction is the dominant hydrogen-burning process in stars similar to, or less massive than, the Sun. See also *fusion*, *hydrogen burning*, *neutron*, *proton*.

protoplanetary disk A flattened disk of dust and gas surrounding a newly formed star and within which matter may be aggregating together to form the precursors of planets. See also *planetesimal*.

protostar A star in the early stages of formation. It consists of the central part of a collapsing cloud that is heating up and is accreting matter from its surroundings, but within which hydrogen fusion reactions have not yet commenced.

pulsar A rapidly rotating neutron star from which we receive brief pulses of radiation, at short and precisely timed intervals, as it spins around its axis.

pulsating variable see *variable star*.

Q

quantum see *photon*.

quark A fundamental particle, the main matter constituent of all atomic nuclei. Quarks join in bunches of three to make baryons (for example, protons and neutrons) or in quark–antiquark pairs to form particles called mesons. See also *antiparticle*, *baryon*.

quasar A very compact but extremely powerful source of radiation that is almost starlike in appearance but is

believed to be the most luminous kind of active galactic nucleus. The name is an abbreviation for quasi-stellar radio source, but is also applied to **quasi-stellar objects (QSOs)**, which are not strong radio emitters.

quiescent prominence see *prominence*.

R

radial velocity The component of a body's velocity that is along the line of sight directly toward, or away from, an observer. The radial velocity of a celestial body can be obtained by measuring the Doppler effect in its spectrum. See also *Doppler effect*, *red shift*, *spectrum*.

radiant The point in the sky from which the tracks of meteors that are members of a particular meteor shower appear to radiate. See also *meteor*.

radio galaxy A galaxy that is exceptionally luminous at radio wavelengths. A typical radio galaxy contains an active galactic nucleus from which jets of energetic charged particles are being propelled toward huge clouds of radio-emitting material that in many cases are much larger than the visible galaxy. See also *active galaxy*.

radio telescope An instrument that is designed to detect radio waves from astronomical sources. The most familiar type is a concave dish that collects radio waves and focuses them onto a detector.

red dwarf star A cool, red, low-luminosity star that, when plotted on a Hertzsprung–Russell diagram, is located toward the bottom end of the main sequence. See also *Hertzsprung–Russell diagram*, *main sequence*.

red-giant star A large, highly luminous star with a low surface temperature and a reddish color. A red giant has evolved away from the main sequence, is “burning” helium in its core rather than hydrogen, and is approaching the final stages of its life. See also *helium burning*, *Hertzsprung–Russell diagram*, *main sequence*.

red shift The displacement of spectral lines to longer wavelengths that is observed when a light source is receding from an observer. The shift in wavelength is proportional to the speed at which the source is receding. **Cosmological red shift** is a wavelength shift that is caused by the expansion of the universe. See also *blue shift*, *Doppler effect*, *spectral line*.

red supergiant star An extremely large star of very high luminosity and low surface temperature. Stars of this kind are located toward the top-right corner of the Hertzsprung–Russell diagram. See also *Hertzsprung–Russell diagram*.